Push Button Simulation

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Introduction

This C++ program simulates a basic finite-state push-button system. The system has two possible states: Off and On. It begins in the Off state and uses user input to determine whether the button has been pressed. The project demonstrates state representation, transition logic, input validation, and a continuous simulation loop.

Planned Test Cases

Test ID	Starting State	Input	Expected Result
TC-01	Program start	None	The initial state is displayed as Off.
TC-02	Off	1	The system transitions to On.
TC-03	On	1	The system remains On.
TC-04	Off	0	The system remains Off.

Test ID	Starting State	Input	Expected Result
TC-05	On	0	The system remains On.
TC-06	Either state	Invalid number	The program rejects the input without crashing.
TC-07	Either state	Letter or symbol	The program rejects the input without entering an infinite error loop.
TC-08	Either state	Exit command	The program terminates normally.

Test Environment

The code was written in C++ and compiled in g++ code space. This was done on a Windows computer with the source file being main.cpp.

Test Results

Test ID	Actual Result	Status
TC-01	Program displayed the initial state as Off.	Pass

Test ID	Actual Result	Status
TC-02	Input 1 transitioned the system from Off to On.	Pass
TC-03	Input 1 while On kept the system On.	Pass
TC-04	Input 0 while Off kept the system Off.	Pass
TC-05	Input 0 while On kept the system On.	Pass
TC-06	Invalid numeric input 7 displayed an error and the program continued.	Pass
TC-07	Invalid text input hello displayed an error and the program continued.	Pass
TC-08	Input q ended the program normally and displayed the final press count.	Pass

Test Summary

All eight planned test cases passed. The program compiled successfully using g++ in GitHub Codespaces. No functional defects were identified during testing.